



Indian Institute of Information Technology, Sri City, Chittoor, Andhra Pradesh, India.

MID EXAMINATION

Sem: 4th (B.Tech.)

Course Name: Fundamentals of Communication

Course Code: EC0204

Time: 1 $\frac{1}{2}$ Hours

Date: 25/02/2026

Max. Marks: 30

Name: [REDACTED] Roll no.: [REDACTED]

Instruction:

- A. All questions are compulsory
- B. Each question carries equal marks.
- C. If required, consider the necessary assumptions.
- D. A scientific calculator is allowed.
- E. Write answers to the questions in one place.

Q. No	Description of Questions		Marks
1.	a)	Define a communication system and list its basic elements?	1
	b)	Write the modulation properties?	1
	c)	Sketch the block diagram of a communication system and discuss the function of each block?	4
2.	a)	What is shot noise? Where does it occur?	1
	b)	Three cascaded amplifiers of figure of merits and gains are given as $F_1 = 2$, $G_1 = 20$, $F_2 = 4$, $G_2 = 15$, $F_3 = 6$. Calculate the total noise figure? Also, identify which stage contributes most to the noise?	2
	c)	Derive the noise figure of cascaded amplifiers?	3
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3.	a)	How do we remove the image station at the superheterodyne receiver?	1
	b)	Compare tuned radio-frequency and Superheterodyne receivers?	2
	c)	Explain the block diagram and working of a low-level AM transmitter?	3
4.	a)	Write two advantages of a superheterodyne receiver.	1
	b)	Explain the selectivity and sensitivity of a receiver?	2
	c)	A superheterodyne receiver is tuned to receive a radio station at 900 kHz. An undesired signal at 1820 kHz is found to interfere due to the	3



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		image frequency. Assume the receiver uses high-side injection and the RF amplifier has a quality factor (Q) of 80. (i) Determine the intermediate frequency (IF) of the receiver? (ii) Calculate the required local oscillator frequency? (iii) Verify whether the given interfering signal (1820 kHz) is actually the image frequency? (iv) Calculate the image rejection ratio (IRR) of the receiver?	
5.	a)	Why is the modulation important in communication?	1
	b)	With neat sketches, illustrate the amplitude-modulated (AM) signal for: (i) Under modulation, (ii) Critical modulation, and Over modulation? Explain how the modulation index affects the envelope of the AM signal?	2
	c)	A carrier of amplitude 50 V is amplitude modulated by a sinusoidal signal of amplitude 30 V. Calculate: (a) Modulation index; (b) Total power; (c) Sideband power, and (d) Transmission efficiency?	3

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